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MILITARY SPECIFICATION

CONTROL GROUP, COMMUNICATION SECURITY
INTEGRATION AND HOUSING GROUP COMPONENTS

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the Integration and Housing Group (IHG) components of the Communications Security Control Group hereinafter referred to as the IHG.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-596	Aluminum Alloy Permanent And Semipermanent Mold Castings

MILITARY

MIL-C-6021	Casting, Classification And Inspection Of
MIL-P-55110	Printed-Wiring Boards
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts: Packaging And Packing Of
MIL-C-28794	Cabinet, Electronic Equipment (Single Bay)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-810C	Environmental Test Methods
MIL-STD-965	Parts Control Program
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)

2.1.2 Other Government publications. The following Government publications form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5810

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PUBLICATIONS

NATIONAL SECURITY AGENCY

NSA-2	National Security Agency Specification For Nameplate And Marking Of Electronic And Electromechanical Equipment
KAG-30A/TSEC	Compromising Emanations Standards For Cryptographic Equipments
NSA 68-8	Multilayer Printed Wiring Boards (Plated Through Holes)
NACSEM 5100 (March 1974)	Compromising Emanations Laboratory Test Standard, Electromagnetic

NAVAL AIR SYSTEMS COMMAND

WS-6536D	Process Specification Procedures And Requirements For Preparation And Soldering Of Electrical Connectors

NAVAL MATERIAL COMMAND

NAVMAT P-9492	Navy Manufacturing Screening Program

NAVAL SEA SYSTEMS COMMAND

S9407-AB-HBK-010	Handbook Of Shipboard Electromagnetic Shielding Practices

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Supplement No. 7 to ARINC SPECIFICATION No. 404 (1956)	Air Transport Equipment Cases And Racking

(Applications for copies should be addressed to ARINC, 2551 Riva Road, Annapolis, MD 21401.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The IHG equipment produced in accordance with this specification shall be physically, functionally, and electrically interchangeable with the IHG equipment produced in accordance with the previous edition of this specification. The IHG shall be in accordance with MIL-E-16400 to the extent specified herein. The IHG shall consist of four ship shelf assemblies (SSAs) specified in a through d:

a. HF-1 SSA
b. HNF-2A SSA
c. HNF-3-1 SSA
d. KG-84 SSA

The SSAs provide interconnection and support for various communications security devices and ancillary equipment. Whenever HNF-1, HNF-2A and HNF-3 are referenced without SSA, HNF-1, HNF-2A and HNF-3 refer to the interconnection boxes specified in e through g:

e. HNF-1 interconnection box
f. HNF-2A interconnection box
g. HNF-3 interconnection box

3.1.1 Standard samples. Standard samples of the components specified in 3.1.2, except n, shall be provided as Government-furnished equipment (GFE). In addition, standard samples of the HNF-1 SSA, HNF-2A SSA, HNF-3 SSA, and KG-84 SSA shall also be provided as GFE.

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3.1.2 SSAs. The SSAs shall be assembled, depending on the required installation, with the components specified in a through n:

a. Mounting base, MT-4841/U
b. Interconnection box HNF-1
c. Interconnection box HNF-2A
d. Interconnection box HNF-3
e. Installation kit, MK-1955/USC
f. KG-84 ship shelf, back panel assembly
g. Remote control unit (RCU) bracket assembly
h. Retainer, ATR box
i. Stud and grounding strap assembly
j. Rails (long and short)
k. Mounting bracket assembly
l. Angle, top support
m. Angle, bottom support
n. Other hardware as required so that the SSAs shall be physically, functionally and electrically interchangeable with the standard sample (see 3.1.1).

Components shall not be welded or glued together; components shall be secured to one another in the same manner as the components are secured together on the standard sample HNF-1 SSA, HNF-2A SSA, HNF-3-1 SSA, and KG-84 SSA.

3.1.3 First article. When specified, samples shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Characteristics. Characteristics of the IHG shall be as specified in 3.2.1 through 3.2.4.

3.2.1 General requirements. IHG components specified in 3.2.1.1 through 3.2.1.12.4 shall be physically, functionally, and electrically interchangeable with the IHG standard samples (see 3.1.1).

3.2.1.1 Mounting base, MT-4841/U. The MT-4841/U mounting base, when combined with the appropriate interconnection box, shall provide the structural support required for mounting the equipment combinations shown in TABLE I. All production MT-4841/U mounting bases shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1). Overall dimensions and weight for the MT-4841/U shall be as specified in TABLE II. In addition, the MT-4841/U shall be mountable in a cabinet built in accordance with MIL-C-28794.

3.2.1.2 Interconnection boxes HNF-1, HNF-2A, and HNF-3. Overall dimensions and weights for the interconnection boxes shall be as specified in TABLE II. All multilayer printed wiring boards (PWBs) utilized in the interconnection boxes shall conform to NSA 68-8. Two-sided PWBs utilized in the interconnection boxes shall conform to MIL-P-55110. Castings utilized in interconnection boxes HNF-1, HNF-2A, and HNF-3 shall conform to requirements of MIL-C-6021, Class 4. The base metal used in the castings shall be aluminum alloy in accordance with QQ-A-596, 356-T51. The finish shall consist of coatings specified in a through d in order of application:

a. Zinc coating applied by the zincate process
b. Electroless nickel applied directly to the zinc; the composition shall be 0.0005 inch (in.) minimum
c. Electrodeposited cadmium 0.0002 in. minimum
d. The cadmium plating shall be given a supplemental chromate treatment

3.2.1.2.1 Interconnection box HNF-1. HNF-1 interconnections shall be as specified in TABLE III. Connectors and switches utilized in the HNF-1 shall be as specified in TABLE IV. Mating connectors and backshells shall be supplied with the HNF-1, and shall be as specified in TABLE V. Contacts used for E1, E2, E3, E4, E5, and E6 shall be compatible with the mark select plug specified in TABLE IV. When viewed from the side of the PWB on which J1, J2, J3, J4, J5, and J6 are mounted, E1, E2, E3, E4, E5, and E6 shall be arranged as shown in FIGURE 1. The HNF-1 shall be configured to permit direct plug-in of two TSEC/KW-46s. All production HNF-1 interconnection boxes shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1). FIGURE 2 shows the general location of HNF-1 connectors.

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3.2.1.2.2 Interconnection box HNF-2A. HNF-2A interconnections shall be as specified in TABLE VI. Connectors and switches utilized in the HNF-2A shall be as specified in TABLE VII. Mating connectors and backshells shall be supplied with the HNF-2A, and shall be as specified in TABLE VIII. The HNF-2A shall be configured to permit direct plug-in of one, two, or three HYX-58/TSEC. All production HNF-2A interconnection boxes shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1). FIGURE 3 shows the general location of HNF-2A connectors.

3.2.1.2.3 Interconnection box HNF-3. HNF-3 interconnections shall be as specified in TABLE IX. Connectors utilized in the HNF-3 shall be as specified in TABLE X. Mating connectors and backshells shall be supplied with the HNF-3, and shall be as specified in TABLE XI. The HNF-3 shall be configured to provide electrical interface for either two TSEC/KY-75s or one TSEC/KY-75 and one HYX-58/TSEC interface unit. All production HNF-3 interconnection boxes shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1). FIGURE 4 shows the general location of HNF-3 connectors.

3.2.1.3 Interconnection cables. The installation kit, MK-1955/USC shall consist of two cables referred to as cable Type IA and cable Type IB. Interconnections for cable Type IA shall be as specified in TABLE XII and interconnections for cable Type IB shall be as specified in TABLE XIII. Connectors utilized in cable Type IA and Type IB shall be as specified in TABLE XIV. MK-1955/USC shall provide the IHG external interconnection between the HNF-3 and the KY-75. The MK-1955/USC shall be physically secured between the HNF-3 and KY-75 with connectors. All production MK-1955/USC installation kits shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1). FIGURE 5 shows connector locations for cable Type IA and Type IB.

3.2.1.4 KG-84 ship shelf, back panel assembly. The KG-84 ship shelf back panel assembly, when used in conjunction with the MT-4841/U, shall enable two KG-84s to be mounted. The back panel assembly shall not inhibit the attachment of cables to either KG-84. All KG-84 ship shelf back panel assemblies shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1).

3.2.1.5 RCU bracket assembly. The RCU bracket assembly shall enable two Z-AKR/TSEC RCUs to be mounted. The RCU bracket assembly shall not inhibit the interconnection via cables of either TSEC/KY-75 to either RCU. The RCU bracket assembly shall be mountable to the MT-4841/U. All RCU bracket assemblies shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1).

3.2.1.6 Retainer, ATR box. ATR box retainers shall conform to ARINC 404, Supplement 7. Matrials used in the construction of the retainer shall conform to the environmental service conditions specified in 3.2.2.1.2. A suggested source of supply for the ATR box retainer is specified in 6.5. All ATR box retainers shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1).

3.2.1.7 Stud and grounding strap assembly. Stud and grounding strap assemblies shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1).

3.2.1.8 Rail. Two types of rails shall be provided; long rails and short rails. The long rails and short rails shall be physically, functionally, and electrically interchangeable with the standard samples (see 3.1.1).

3.2.1.9 Mounting bracket assembly. Mounting bracket assemblies shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1).

3.2.1.10 Angle, top support. Top support angles shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1).

3.2.1.11 Angle, bottom support. Bottom support angles shall be physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1).

3.2.1.12 SSAs. The SSAs, specified in 3.1, shall be in accordance with 3.2.1.12.1 through 3.2.1.12.4. In addition, each SSA shall be mountable in a cabinet built in accordance with MIL-C-28794.

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3.2.1.12.1 HNF-1 SSA. The HNF-1 SSA shall provide all mechanical and electrical interfaces required to interconnect and mount two TSEC/KW-46 equipments and shall combine the MT-4841/U mounting base, the HNF-1 interconnection box, ATR box retainers, rails, and the various other hardware required to make the assembly physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1).

3.2.1.12.2 HNF-2A SSA. The HNF-2A SSA shall provide all mechanical and electrical interfaces required to interconnect and mount three HYX-58/TSEC equipments and shall combine the MT4841/U mounting base, the HNF-2A interconnection box, ATR box retainers, rails, and the various other hardware required to make the assembly physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1).

3.2.1.12.3 HNF-3-1 SSA. The HNF-3-1 SSA shall provide all mechanical and electrical interfaces required to interconnect and mount two TSEC/KY-75 equipments and two Z-AKR/TSEC RCUs. The assembly shall combine the MT-4841/U mounting base, the HNF-3 interconnection box, two MK-1955/USC installation kits, the RCU bracket assembly, two mounting bracket assemblies, two stud and grounding strap assemblies, ATR box retainers, rails, and the various other hardware required to make the assembly physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1).

3.2.1.12.4 KG-84 SSA. The KG-84 SSA shall provide all mechanical and electrical interfaces required to mount two TSEC/KG-84 equipments. The assembly shall combine the MT-4841/U mounting base, the KG-84 ship shelf back panel assembly, ATR box retainers, rails, and the various other hardware required to make the assembly physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1).

3.2.2 Detailed requirements. Detailed requirements shall be in accordance with 3.2.2.1 through 3.2.2.2.

3.2.2.1 Service conditions (general). The IHG shall perform with all voltages, loads, and input/output (I/O) signals at their worst case; and to the electrical and environmental conditions specified in 3.2.2.1.1 through 3.2.2.1.2.6.

3.2.2.1.1 Service conditions (electrical). Electrical service conditions shall be in accordance with 3.2.2.1.1.1 through 3.2.2.1.1.5.

3.2.2.1.1.1 Electrical power characteristics. The HNF-1, HNF-2A, HNF-3, and MK-1955/USC shall operate in accordance with the shipboard requirements specified in MIL-E-16400 and DoD-STD-1399, Section 300 within the limits specified in a and b:

a. Operating voltage (nominal conditions):
   1. 115 volts alternating current (VAC) ±8.05 VAC at 60 Hertz (Hz) ±1.8 Hz
   2. 115 VAC ±8.05 VAC at 400 Hz ±20 Hz
b. Operating frequency range shall be as specified in a.

3.2.2.1.1.2 Electrical performance. The IHG shall provide the required circuitry to conform to the requirements specified in 3.2.1 through 3.2.1.12.4. The IHG shall perform with all voltages, loads, and I/O signals at their worst case, and to the electrical conditions specified in 3.2.2.1.1.1 and the environmental conditions specified in 3.2.2.1.2 to 3.2.2.1.2.6.

3.2.2.1.1.3 TEMPEST. The HNF-1 and HNF-2A interconnection boxes shall conform to the compromising emanations requirements of KAG-30A/TSEC and NACSEM 5100. The HNF-3 interconnection box and installation kit MK-1955/USC shall conform to the compromising emanations requirements of NACSEM 5100.

3.2.2.1.1.4 Electromagnetic interference (EMI) and electromagnetic compatibility (EMC). Insofar as they do not conflict with the interchangeability requirements of this specification, the designation and marking, separation, spacing, and shielding requirements of the Cable Spacing And Shielding Requirements section of S9407-AB-HBK-010 shall apply to the IHG interconnecting cables.

3.2.2.1.1.5 Bonding and grounding. The Bonding and grounding paragraph of MIL-E-16400 shall apply to the IHG.

3.2.2.1.2 Service conditions (environmental). The IHG shall be capable of reliable operation under the environmental conditions specified in 3.2.2.1.2.1 through 3.2.2.1.2.6. Whenever MIL-E-16400 references MIL-STD-810, MIL-STD-810C shall be invoked.

3.2.2.1.2.1 Temperature. The IHG shall conform to Range 4 temperature requirements of MIL-E-16400.

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3.2.2.1.2.2 Vibration. The IHG shall conform to the Environmental vibration paragraph of MIL-E-16400 for shipboard equipment.

3.2.2.1.2.3 Shock. The IHG shall conform to the Shock paragraph of MIL-E-16400.

3.2.2.1.2.4 Humidity. The IHG shall conform to the Humidity paragraph of MIL-E-16400.

3.2.2.1.2.5 Salt fog. The IHG shall conform to the Salt fog paragraph of MIL-E-16400 for sheltered equipment. After the salt fog test, the IHG shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface exfoliation. There shall be no clogging or binding or moving parts.

3.2.2.1.2.6 Fungus. The IHG shall be resistant to fungi in both operating and nonoperating conditions. After the fungus test, the IHG, when examined with the aid of a 10-power magnifier shall show no more than six minute, unrelated spots, each no greater than 0.015 square inch (sq in.) in area, of sparse microbial growth as evidenced by growth colonization (which includes branching) and sporulation on or within each cubic foot (cu ft), or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination shall not be included in this evaluation.

3.2.2.2 Enclosures. The HNF-1, HNF-2A, and HNF-3 shall conform to the Degree of enclosure paragraph of MIL-E-16400 (15 degrees dripproof).

3.2.3 Safety criteria. The IHG shall conform to the Safety criteria paragraph of MIL-E-16400.

3.2.4 Dimensions and weight. The dimensions and weight of all IHG components shall be such that they are physically, functionally, and electrically interchangeable with the standard sample (see 3.1.1). In addition, for those components listed in TABLE II, dimensions and weight shall be as shown in TABLE II.

3.3 Parts, materials, and processes. Parts, materials and processes for the IHG shall be in accordance with the Parts, materials, and processes paragraph of MIL-E-16400, except as specified in 3.3.1 through 3.3.4. In the event of conflict between any other part of this specification with the Parts, materials, and processes paragraph of MIL-E-16400, this specification shall take precedence.

3.3.1 Parts control. The parts to be incorporated in the IHG shall be controlled in accordance with MIL-STD-965, Procedure I.

3.3.2 Connectors. Connectors utilized in the IHG shall be as specified in TABLE IV, TABLE VII, TABLE X, and TABLE XIV.

3.3.3 Unacceptable materials. The Unacceptable materials paragraph of MIL-E-16400 shall apply. The IHG design shall not include the materials specified in a through f:

a. Polychlorinated biphenyls
b. Asbestos and asbestos compounds
c. Flammable or combustible materials
d. Fragile or brittle materials
e. Beryllium and beryllium compounds, unless so identified
f. Lithium and lithium compounds, unless specifically approved by the procuring activity

The use of mercury-containing components is forbidden.

3.3.4 Soldering. Soldering shall be in accordance with WS-6536D; however, PWBs shall be as specified in 3.2.1.2.

3.4 Identification and marking. Identification and marking shall be as shown on the standard samples (see 3.1.1), except that the nameplates for the HNF-1, HNF-2A, and HNF-3 shall be in accordance with Group 1, Type 1, Class 4 requirements of NSA-2. FIGURE 6 may be used as a guide for the nameplate appearance and dimensions for the HNF-1, HNF-2A, and HNF-3; however, the final nameplates are subject to Government approval.

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3.5 Workmanship. Workmanship shall be in accordance with 3.5.1 and 3.5.2.

3.5.1 General workmanship. Workmanship of the IHG shall be in accordance with the Workmanship paragraph of MIL-E-16400.

3.5.2 Workmanship screen. The IHG shall withstand a defect detection vibration screen of random type vibration at 0.04 g^2/Hz ±3 decibels (dB) from 80 Hz to 350 Hz and temperature cycling specified in 4.7.2.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited, to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied; b) Government product inspection to measure quality of the product to be offered for acceptance; and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through d:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
c. Safety (see 4.5)
d. Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.3), the minimum equipment required for first article inspection shall be as specified in a through d:

a. Two HNF-1 SSA
b. Two HNF-2A SSA
c. Two HNF-3-1 SSA
d. Two KG-84 SSA

First article inspection shall consist of all examinations and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE XV.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4.

4.4.1 Production inspection (Group A). Production inspection shall be made on every IHG offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process such as limited functional and performance tests, tests which detect deviations from design, and tests of controls and adjustments. Production inspection shall include the examinations and tests specified in TABLE XVI.

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4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. Production control inspection shall encompass tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs; tests which detect deviatons in the processing of the materials; and tests of the performance in a system. These tests shall be performed on equipments that have passed production inspection. Production control inspection shall include the examinations and tests specified in 4.6.3.5, 4.6.4, and 4.6.5. The equipment shall satisfactorily conform to the requirements of production control inspection prior to releasing the lot for shipment.

4.4.2.1 Sampling plan for production control inspection. IHG equipment, selected for production control inspection, shall be selected by the contractor under the supervision of the Government Quality Assurance Representative (QAR) and shall be representative of current production. TABLE XVII shall be used to determine sample composition for the sample sizes specified. Sampling shall conform to the procedures of small-sample inspection of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection, and the acceptable quality level (AQL) shall be 6.5 percent. The lot or batch size range of TABLE I of MIL-STD-105 that now reads 26 to 50 shall be changed to 26 to 75 and the lot or batch size range that now reads 51 to 90 shall be changed to 76 to 90.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot, using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be accomplished on a sampling basis as specified in 4.4.3.1. Environmental inspection shall include the examinations and tests specified in TABLE XVIII. Group C inspection shall be performed on units that have passed Group A and Group B inspections. Group C inspection shall be performed in an order acceptable to the Government.

4.4.3.1 Sampling plan for Group C inspection. All samples shall include an HNF-1 SSA, an HNF-2A SSA, and an HNF-3 SSA. The first sample shall be selected from the first month's production. The second sample shall be selected from the seventh month's production. Samples shall continue to be taken at seven months intervals from the months production for which the sample is taken until production ceases.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the QAR the results thereof and details of the corrective action taken to correct units of production which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure the matter shall be referred to the contracting officer (see 6.4).

4.4.4 Reinspection of conforming production control inspection and environmental inspection sample units. Sample units which have been subjected to, and have passed production control or environmental inspections, may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.5 Safety inspection. The IHG shall be visually examined to assure the absence of sharp corners or projections that could cause injury, directly or indirectly, by catching onto clothing, to verify compliance with 3.2.3. If the contractor makes any changes to materials, processes, or the design after the first unit is accepted, the contracting officer may require a new inspection of a unit produced under the new conditions. If a unit fails the safety inspection, the contractor shall immediately investigate the cause of failure and report to the procuring activity the results and actions taken to correct units which were manufactured under the same conditions. No unit produced under a set of conditions shall be accepted until a unit, produced under those conditions, has passed the safety inspection.

4.6 Test methods. Unless otherwise specified in 4.6.1 through 4.6.6, the inspections specified in TABLE XV, TABLE XVI, and TABLE XVIII, as well as 4.6.1, shall be performed in accordance with the test methods specified in MIL-E-16400. Whenever MIL-E-16400 references MIL-STD-810, MIL-STD-810C, shall be invoked. Operating tests, when required, shall consist of the connectivity check specified in 4.6.1.2, except when required during vibration or shock tests. Operating tests required during vibration or shock tests shall consist of observing that equipment mounted in an SSA remains energized. When an SSA is required to be mounted for other than interchangeability check 2 of TABLE XIX, the SSA shall be mounted as shown in FIGURE 7 using stand-alone mounting brackets. The values specified in a through c shall be used as a basis for room ambient

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conditions to establish normal performance levels for testing:

a. Temperature, room ambient 23° Celsius (C) ±5°C
b. Humidity, 50 percent ±30 percent relative
c. Atmospheric pressure, 28.5 in. of mercury (in. Hg) +2.0 in. Hg to -4.5 in. Hg

4.6.1 Functional tests. Functional tests specified in 4.6.1.1 and 4.6.1.2 shall be performed to verify that IHG components are physically, functionally, and electrically interchangeable with sample IHG components.

4.6.1.1 Interchangeability. The tests specified in TABLE XIX, TABLE XX, TABLE XXI, TABLE XXII, and TABLE XXIII shall be performed to verify interchangeability. FIGURE 8 shows the type and mounting positions of equipments used to populate the SSA under test for those tests specified in TABLE XIX. Production SSAs shall be used for the tests specified in TABLE XIX. The tests specified in TABLE XX shall be performed using the sample mating connectors provided by the Government and production SSAs. The tests specified in TABLE XXI shall be performed using the sample MK-1955/USC installation kit provided by the Government and a production HNF-3-1 SSA; however, prior to the test, the production MK-1955/USC installation kits shall be removed from the production HNF-3-1 SSA. The tests specified in TABLE XXII shall be performed using the sample MK-1955/USC provided by the Government, a production MK-1955/USC, and the sample TSEC/KY-75 provided by the Government. The tests specified in TABLE XXIII shall be performed using a production MK-1955/USC and the sample HNF-3-1 SSA provided by the Government; however, prior to the tests the sample MK-1955/USC installation kits shall be removed from the sample HNF-3-1 SSA. IHG components not conforming to the requirements in TABLE XIX, TABLE XX, TABLE XXI, TABLE XXII, and TABLE XXIII shall be considered as having failed the interchangeability requirements.

4.6.1.2 Connectivity. Continuity between interconnections specified in TABLE III, TABLE VI, TABLE IX, TABLE XII, and TABLE XIII shall be verified for the HNF-1, HNF-2A, HNF-3, and MK-1955/USC, respectively. If continuity does not exist, the IHG components shall be considered as having failed.

4.6.2 TEMPEST. HNF-1, HNF-2A, and HNF-3 interconnection boxes and MK-1955/USC installation kit shall be subjected to TEMPEST tests according to a test plan approved by the procuring activity (see 6.2.2) to verify compliance with 3.2.2.1.1.3.

4.6.3 Environmental tests. Environmental tests shall be in accordance with 4.6.3.1 through 4.6.3.6.

4.6.3.1 Temperature. The IHG shall be subjected to the tests specified in 4.6.3.1.1 and 4.6.3.1.2 to verify compliance with 3.2.2.1.2.1.

4.6.3.1.1 Low temperature. The IHG shall be subjected to the low temperature tests of MIL-E-16400.

4.6.3.1.2 High temperature. The IHG shall be subjected to the high temperature tests of MIL-E-16400.

4.6.3.2 Vibration. The IHG shall be subjected to the environmental vibration test of MIL-E-16400 to verify compliance with 3.2.2.1.2.2.

4.6.3.3 Shock. The IHG shall be subjected to the shock test of MIL-E-16400 for lightweight shipboard equipment to verify compliance with 3.2.2.1.2.3.

4.6.3.4 Humidity. The IHG shall be subjected to the humidity test of MIL-E-16400 to verify compliance with 3.2.2.1.2.4.

4.6.3.5 Salt fog. The IHG shall be subjected to the salt fog tests for sheltered equipment of MIL-E-16400 to verify compliance with 3.2.2.1.2.5. When this test is performed on the HNF-1 SSA, HNF-2A SSA, or HNF-3-1 SSA the interconnection box shall be removed from the SSA. After removal, the interconnection box shall be disassembled and the gaskets and printed wiring assembly removed. The disassembled interconnection box without gaskets or printed wiring assembly shall be subjected to the salt fog test. The functional test specified in 4.6.1.2 shall be conducted prior to, and after, the salt fog test. Failure criteria shall be limited to corrosion of finishes and metals and functional test pass or fail criteria. Corrosion of finishes and metals shall be defined as any visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

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4.6.3.6 Fungus. The IGH shall be subjected to the fungus test of MIL-E-16400 to verify compliance with 3.2.2.1.2.6. The functional test of 4.6.1.2 shall be performed prior to, and after, the fungus test. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling prior to, and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to conform to fungus requirements specified in 3.2.2.1.2.6 and the functional tests specified in 4.6.1.2 shall consistute failure of this test.

4.6.4 Enclosure. The HNF-1, HNF-2A, and HNF-3 shall be subjected to the enclosure tests of MIL-E-16400 to verify compliance with 3.2.2.2.

4.6.5 Surface examination. The SSAs shall be examined to verify conformance with the examination specified in a through s:

a. Dimension and weight (see 3.2.4)
b. Parts, materials, and processes (see 3.3)
c. Identification and marking (see 3.4)
d. Finish (see 3.2.1.2)
e. Assembly (see 3.1.2)
f. Bonding and grounding (see 3.2.2.1.1.5)
g. EMI and EMC (see 3.2.2.1.1.4)
h. Completeness
i. Operation of adjustable and sliding parts, thumb screws, controls, and switches
j. Welded and soldered joints
k. Fit of components in their respective positions
l. Check of mounting means
m. Lubrication and rust prevention
n. Safety measures and interlocks
o. Fastening and securing devices or parts
p. Cable runs between components, including plugs and receptacles
q. Grounding connections
r. Foreign objects, lint, and dust
s. Potential shorts and defective insulation

In performing the examination, the SSAs shall not be disassembled in a manner that will effect the performance, durability, or appearance of the SSAs.

4.6.6 Electrical power tests. The alternating current (AC) power inputs of the HNF-1, HNF-2A, and HNF-3 shall be subjected to the Spike voltage test of MIL-E-16400 to ensure compliance with 3.2.2.1.1.1. The spike voltage test shall be performed for both positive and negative spikes. Following the test, a continuity check shall be performed on the AC inputs. The AC inputs shall be checked to ensure that the inputs are not shorted to one another or to ground.

4.7 Workmanship. Workmanship shall be as specified in 4.7.1 through 4.7.2.2.

4.7.1 General workmanship. The IHG shall be examined for workmanship for conformance to the requirements specified in 3.5.1.

4.7.2 Workmanship screen. Using NAVMAT P-9492 for guidance, the workmanship screen shall consist of random vibration and temperature cycling specified in 4.7.2.1 and 4.7.2.2. The screening may be performed at the shelf, interconnection box, or end item level and shall be performed on each IHG equipment including cables and connectors; however, the random vibration screen need not be performed on the MT-4841/U mount.

4.7.2.1 Random vibration screen. Vibration shall be performed prior to conducting the temperature cycling test. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes on the axis deemed most susceptible to vibration excitation. All IHG equipment shall be hardmounted (without shock isolators) and subjected to the vibration conditions of FIGURE 9. Input vibration levels shall be measured at the mounting points of the IHG under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. The instantaneous acceleration peaks may be limited to three times root-mean-square (rms) acceleration level. Equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD). The connectivity test of 4.6.1.2 shall be performed during vibration to observe any abnormal conditions. Any failure occurring during screening shall require corrective action before the vibration is resumed.

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4.7.2.2 Temperature cycling. Each IHG equipment shall be subjected to 3 cycles of the temperature curve shown in FIGURE 10. The temperature rate of change shall not be less than 5°C per minute. IHG equipments shall not have AC power applied during these tests. The IHG equipment shall be positioned for maximum exposure to changing temperature. Where performance measurements are called for in FIGURE 10, the connectivity test specified in 4.6.1.2 shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the IHG equipment reaches 80 percent of the chamber temperature. When failures occur, the IHG equipment shall be reworked and the cycling continued for a cumulative total of 3 cycles.

4.8 Inspection of preparation for delivery. Inspections shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The IHG equipment covered by this specification provides interconnection and support for various communications security equipment and is intended for use on-board ships.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify:

a. Title, number, and date of this specification.
b. Levels of preservation, packaging, packing, and marking required (see 5.1).
c. Number of first article samples to be submitted if other than specified in 4.3.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraphs	Data requirement	Applicable DID
3.3.1	Non-standard Part Approval Requests/ Proposed Additions To An Approved Program Parts Selection List	DI-E-7028A May 1981
3.3.1	Military Detail Specifications And Specification Sheets	DI-E-7029 30 March 1977
3.3.1	Test Data For Non-standard Parts	DI-E-7030 30 March 1977
3.2.2.1.1.3	TEMPEST Test Set-up Ambient Signal Control Certification Report, Electromagnetics	DI-T-5183A 26 October 1973
3.2.2.1.1.3	TEMPEST Detection System Certification Report, Electromagnetics	DI-T-5182A 26 October 1973

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Paragraphs	Data requirement	Applicable DID
3.2.2.1.1.3	TEMPEST Test Facility Certification Report, Electromagnetics	DI-T-5181A 26 October 1973
3.2.2.1.2	Test Plan	DI-T-5204
3.1.3	First Article Qualification Test Plan	DI-T-5315A 12 May 1977
4.4	Quality Status Report Of Production Testing	DI-T-5156C 29 March 1974
4.6	Test Report	DI-T-5247 19 April 1976
4.6	Procedures, Test	UDI-T-22710B 1 July 1976
4.6.2	TEMPEST Test Evaluation Report, Electromagnetics	DI-T-5180A 26 October 1973
4.6.2	TEMPEST Test Plan, Electromagnetics	DI-T-5140B 26 October 1973

(DIDs related to this specification, and identified in Section 6 will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article is required, it should be tested and approved under the appropriate provisions of 52.209-3 of the Federal Acquisition Regulations. The first article should be a first production item. The first article should, as a minimum, consist of the units specified in 4.3. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 Environmental inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action (see 4.4.3.2).

6.5 Suggested source of supply for the ATR box retainer. A suggested source of supply for the ATR box retainer is TRI TECH, FSCM 32302, St. Petersburg, FL 33733.

6.6 GFE. The contracting officer should arrange to furnish the equipment listed in 3.1.1, the mating connectors specified in TABLE V, TABLE VIII, and TABLE XI, and as specified in a through f:

a. TSEC/KY-75 (two)
b. TSEC/KW-46 (three)
c. TSEC/KY-58 (three)
d. TSEC/KG-84( ) (two)
e. HYX-58/TSEC (three)
f. Z-AKR/TSEC RCU and cable (two)

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6.7 Supersession data note. This specification supersedes all previous specifications covering the IHG equipments. The superseded documents are specified in a through h:

a.	MIL-C-28863(EC)	Control Group, Communications Security, Integration And Housing Group Components
b.	ELEX-B-277	Box, Interconnecting J-3423(U)
c.	ELEX-B-278	Base, Mounting, MT-4841(U)
d.	ELEX-C-279	Cable Kit, Standard, OK-313(V)/USC
e.	ELEX-B-308	Box, Interconnecting, J-3423(U)
f.	ELEX-B-309	Base, Mounting, MT-4841(U)
g.	ELEX-C-310	Cable Kit, Standard, OK-313(V)/USC
h.	ELEX-C-353	Control Group, Communications Security Integration And Housing Group Components

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY-EC

(Project No. 5810-N049(EC))

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TABLE I. Equipment combinations.

Quantity	Equipment	Quantity	Equipment	Interconnection box or back panel assembly
3	HYX-58/TSEC/KY-58	and --	--	HNF-2A
2	TSEC/KG-84	and --	--	KG-84 back panel assembly
2	TSEC/KW-46	and --	--	HNF-1
2	TSEC/KY-75	and --	--	HNF-3
1	TSEC/KY-75	and 1	HYX-58/TSEC/KY-58	HNF-3

TABLE II. Physical dimensions and weight. 1/

Nomenclature	Height		Width		Depth		Weight 2/	
	mm 3/	in.	mm	in.	mm	in.	kg 4/	lbs 5/
MT-4841/U	276.2	10.875	482.6	19.00	469.9	18.5	9.1	20
HNF-1	240.1	9.61	435.1	17.13	76.2	3.00	6.81	15
HNF-2	240.1	9.61	435.1	17.13	76.2	3.00	6.81	15
HNF-3	240.1	9.61	166.4	6.55	50.8	2.00	3.64	8
MK-1955/USC	N/A	N/A	N/A	N/A	787.4	31.0	2.27	5.0

1/ Dimensions and weight maximum (weight tolerance +0 percent, -1 percent)
2/ Uncrated
3/ Millimeters
4/ Kilograms
5/ Pounds

TABLE III. HNF-1 interconnections.

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	J9 Pin	J10 Pin	J11 Pin	Other 1/
L						25 2/					
M						29					
b						26					
c						30					
d						27					
R						23					
S						24					
e						28					
G									23		
B									25		
V									26		
F									24		
C									29		
W									30		
X									28		
Y									27		

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TABLE III. HNF-1 interconnections. (continued)

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	J9 Pin	J10 Pin	J11 Pin	Other 1/
T											Ground
A											Ground
U											Ground
Z											Ground
J											Ground
a											Ground
h											Ground
f											Ground
K											Ground
H											Ground
P											Spare
E											Spare
											Spare
											Spare
							12			32	
	f, C						11			31	
	N, J						22				
	g						21				
	R						32			12	
	J, Y						31			11	
	W, B						42				
	S						41				
	T						14				
	i						25				
	U									22	
	Z									21	
	D									42	
	E									41	
	F									14	
	b									25	
	H										Ground
	A										Ground
	X										Ground
	L										Ground
	t										Ground
	q										Ground
	k										Ground
	p										Ground
	d										Ground
	e										Ground
	M										Spare
	K										Spare
	G										Spare
	P										Spare
		A						138			
		B						139			
		C						140			
		E						104			
		F						108			
		K						105			
		J						101			

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TABLE III. HNF-1 interconnections. (continued)

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	J9 Pin	J10 Pin	J11 Pin	Other 1/
		G									Ground
		L									Ground
			C				38	153, 156		38	
			A				30	149, 152		30	
			B				3, 13	157, 158		3, 13	
			B				23, 33	159		23, 33	
			B				43	160		43	
			D								Spare
			E								Spare
				P	P			69			Input FL1
				N	N			73			Input FL2
				M	M			77			Input FL3
				G	G	41		33	41		
				E	E	31		28	31		
				a	a	42		29	42		
				Z	Z	32		27	32		
				Y	Y	20		25	20		
				C	C	21		26	21		
				K	K	1		36	1		
				C	C	2		35	2		
				h	h	10		9	10		
				S	S	11		13	11		
				j	j	8		1	8		
				W	W	7		5	7		
N, D				B, D, F	B, D, F						
N, D				J, T, V	J, T, V						
							6			6	Output FL3
							34			34	Output FL3
							51				Output FL1
						33					S1-1
						34					S1-2
						36					S1-3
						37					S1-4
						45					S1-6
						46					S1-7
						47					S1-8
						44					S1-5
						43					S1-9
						43					S1-10
						43					S1-11
						43					S1-12
						43					S1-13
						43					S1-14
						43					S1-15
						43					S1-16
									36		S2-3
									34		S2-2
		M									Spare
		D									Spare

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TABLE III. HNF-1 interconnections. (continued)

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	J9 Pin	J10 Pin	J11 Pin	Other 1/
										51	Output FL2
									33		S2-1
									37		S2-4
									44		S2-5
									45		S2-6
									46		S2-7
									47		S2-8
									43		S2-9
									43		S2-10
									43		S2-11
									43		S2-12
									43		S2-13
									43		S2-14
									43		S2-15
									43		S2-16
											Spare
											Spare
											Spare
											Spare
				UL	UL						Ground
								87			Ground
								93			Ground
								111			Ground
				Axkr	Axkr						Ground
											Ground
											Ground
											Ground
											Ground
											Ground
									49		J10-48
								88			J9-85
						49					J7-48
								86			Ground
E1 open E2 open E3 to J8-44 E4 to J11-44 E5 to ground E6 to ground											

1/ All points labeled ground are interconnected.
2/ All points in the same horizontal line are interconnected.

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TABLE IV. HNF-1 connectors and switches.

Quantity	Description	Suggested source(s) and part number
1	Connector assembly J9	Collins Radio Company 1/ 619-7850-002, or equal
40	Contacts used with connector assembly J9	Collins Radio Company 327-2605-013, or equal
1	J1	ITT Cannon Electric 2/ KPT2E18-32PDV, or equal
1	J2	ITT Cannon Electric KPT2E20-41PYDV, or equal Burndy Corp. 3/ BTO2E2041PYH145, or equal
1	J3	ITT Cannon Electric KPT2E14-19PWDV, or equal
1	J4	ITT Cannon Electric KPT2E14-5PDV, or equal
2	J5, J6	ITT Cannon Electric KPT2E20-41PWDV, or equal Burndy Corp BTO2E2041PWH145, or equal
4	J7, J8, J10, J11	M83733-04RA051
2	Switches S1 and S2	Amphenol, Inc. 435166-5 4/, or equal Grayhill Inc. 5/ 76SB085, or equal
1	Mark select plug	Circuit Assembly Corporation 6/ CA-3003-002, or equal

1/ Federal Supply Code for Manufacturers (FSCM) 13499
2/ FSCM 71468
3/ FSCM 09922
4/ FSCM 00779
5/ FSCM 81073
6/ FSCM 52072

TABLE V. HNF-1 mating connectors and backshells.

Quantity	Description	Suggested source and part number
1	Backshell	Glenair, Inc. 1/ 390DS006NF1810A6, or equal
1	Backshell	Glenair, Inc. 390DS006NF2012A6, or equal
2	Backshell	Glenair, Inc. 390DS006NF1410A6, or equal
1	Mating connector	MS3126E-18-32S
1	Mating connector	MS3126E-20-41SY
1	Mating connector	MS3126E-14-19SW
1	Mating connector	MS3126E-14-5S

1/ FSCM 06324

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TABLE VI. HNF-2A interconnections.

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	J9 Pin	Other 1/
A						47 2/			
r						58			
B						79			
E						70			
D						60			
Y						57			
C						52			
X						48			
W						46			
q						45			
Z						54			
a						62			
s						63			
CC						59			
u							45		
c							46		
d							48		
J							52		
e							57		
K							60		
L							70		
H							79		
G							58		
Y							62		
g							54		
f							47		
EE							59		
W							63		
k								57	
S								60	
V								45	
1								46	
J								48	
R								52	
T								70	
P								79	
N								58	
z								62	
n								47	
m								54	
p									Ground
BB									Ground
HH									Ground
DD									Ground
FF									Ground
x									Ground
t									Ground
h									Ground
b									Ground
F									Ground
M									Ground
	g					101			
	J					135			
	H					133			
	f					104			
	E					108			
	G					105			
	K					111			
	L					138			

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TABLE VI. HNF-2A interconnections. (Continued)

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	J9 Pin	Other 1/
	i					139			
	j					140			
	h					136			
	AA					110			
	F					86, 87			
	z					137			
	p						136		
	q						140		
	r						139		
	V						138		
	U						111		
	S						133		
	T						135		
	n						101		
	N						108		
	m						104		
	R						105		
	DD						110		
	P						86, 87		
	EE						137		
	u							136	
	t							104	
	X							108	
	Z							101	
	b							135	
	A							133	
	a							105	
	v							110	
	Y							86, 87	
	w							137	
	W								Ground
	s								Ground
	FF								Ground
	NN								Ground
	PP								Ground
	JJ								Ground
	LL								Ground
	BB								Ground
	x								Ground
	e								Ground
	k								Ground
	M								Ground
	D								Ground
	KK								Ground
	MM								Ground
	HH								Ground
	GG								Spare
	CC								Spare
	y								Spare
		G				89			
		E				93			
		F				85			
		L					89		
		J					93		
		K					85		
		A						89	
		C						93	
		B						95	
		P							Spare
		S							Spare

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TABLE VI. HNF-2A interconnections. (Continued)

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	J9 Pin	Other 1/
							152		
			A					149	
			A					152	
			A						Ground
			B			157			Ground
			B			158			Ground
			B			159			Ground
			B			160			Ground
			B				157		Ground
			B				158		Ground
			B				159		Ground
			B				160		Ground
			B					157	Ground
			B					158	Ground
			B					159	Ground
			B					160	Ground
			B			153			
			C			156			
			C				153		
			C				156		
			C					153	
			C					156	
			C						Spare
			D						Spare
			E						
				J	J	1	1	1	
				W	W	5	5	5	
				J	J	79	79	79	
				V	V	79	79	79	
				T	T	79	79	79	
				B	B	79	79	79	
				D	D	79	79	79	
				F	F	79	79	79	
				h	h	9	9	9	
				S	S	13	13	13	
				Y	Y	25	25	25	
				C	C	26	26	26	
				Z	Z	27	27	27	
				E	E	28	28	28	
				a	a	29	29	29	
				G	G	33	33	33	
				c	c	35	35	35	
				K	K	36	36	36	
				A	A				Ground
				X	X				Ground
				k	k				Ground
				t	t				Ground
				r	r				Ground
				g	g				Ground
				R	R				Ground
				q	q				Ground
				p	p				Ground
				e	e				Ground
				d	d				Ground
				L	L				Ground
						109			Ground
						112			Ground
							109		Ground
							112		Ground

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TABLE VI. HNF-2A interconnections. (Continued)

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	J9 Pin	Other 1/
								109	Ground
								112	Ground
						22			J7-23
							22		J8-23
								22	J9-23
						30			S1-1
						31			S1-2
						37			S1-3
						38			S1-4
						39			S1-5
						40			S1-6
						49			S1-7
						50			S1-8
						53			S1-9
						53			S1-10
						53			S1-11
						53			S1-12
						53			S1-13
						53			S1-14
						53			S1-15
						53			S1-16
							30		S2-1
							31		S2-2
							37		S2-3
							38		S2-4
							50		S2-8
							53		S2-9
							53		S2-10
							53		S2-11
							53		S2-12
							53		S2-13
							53		S2-14
							53		S2-15
							53		S2-16
								30	S3-1
								31	S3-2
								37	S3-3
								38	S3-4
								39	S3-5
								40	S3-6
								49	S3-7
								50	S3-8
								53	S3-9
								53	S3-10
								53	S3-11
								53	S3-12
								53	S3-13
								53	S3-14
								53	S3-15
								53	S3-16
							39		S2-5
							40		S2-6
							49		S2-7
				P	P	69	69	69	
				N	N	73	73	73	
				M	M	77	77	77	

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TABLE VI. HNF-2A interconnections. (Continued)

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	J9 Pin	Other 1/
		U							Spare
			A			149			
			A			152			
			A				149		
	B							111	
	C							138	
	d							139	
	c							140	
AA								63	
GG								59	
U									Ground
V									Ground

1/ All points marked ground are interconnected.
2/ All points in the same horizontal line are interconnected.

TABLE VII. HNF-2A connectors and switches.

Quantity	Description	Suggested source and part number
3	Connector assemblies J7, J8, and J9	Collins Radio Company 1/ 619-7850-002, or equal
201	Contacts for use with connector assemblies J7, J8, and J9	Collins Radio Company 372-2605-013, or equal
1	J1	ITT Cannon Electric 2/ KPT2E22-55PDV, or equal Burndy Corporation 3/ BTO2E22-55PH145, or equal
1	J2	ITT Cannon Electric KPT2E24-61PDV, or equal
1	J3	ITT Cannon Electric KPT2E14-19PXDV, or equal
1	J4	ITT Cannon Electric KPT2E14-5PDV, or equal
2	J5, J6	ITT Cannon Electric KPT2E20-41PWDV, or equal Burndy Corporation BTO2E2041PWH145, or equal
3	Switches S1, S2, S3	Amphenol, Incorporated 4/ 435166-5, or equal Grayhill Incorporated 5/ 76SB085, or equal

1/ FSCM 13499
2/ FSCM 71468
3/ FSCM 09922
4/ FSCM 00779
5/ FSCM 81073

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TABLE VIII. HNF-2A mating connectors and backshells.

Quantity	Description	Suggested source and part number
1	Backshell	Glenair, Incorporated 1/ 390DS006NF2216A6, or equal
1	Backshell	Glenair, Incorporated 390DS006NF2416A6, or equal
2	Backshell	Glenair, Incorporated 390DS006NF1410A6, or equal
1	Mating connector	MS3126E-22-55S
1	Mating connector	MS3126E-24-61S
1	Mating connector	MS3126E-14-19SX
1	Mating connector	MS3126E-14-5S

1/ FSCM 06324

TABLE IX. HNF-3 interconnections.

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	Other 1/
n m b a Y B c J H E D C Z G F V T h f N W j U P S q i R r s				K 2/ T f E e S Y G P L N A B C R, F, D, M	P A B e S Y G C N L f E D, M, F, R K T			

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TABLE IX. HNF-3 interconnections. (Continued)

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	Other 1/
A L X e k p t	C J G F K H E B C Y a Z D W J U S R V P f M N T g h h Z A L X d k p t	A B E D F H C G J K	A C E B	U J	J U	Z G J K H L c M N P a d b e S d A, B D, E C C F	Z G J c K H L M N P b a d e S a A, B D, E C	Ground Ground

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TABLE IX. HNF-3 interconnections. (Continued)

J1 Pin	J2 Pin	J3 Pin	J4 Pin	J5 Pin	J6 Pin	J7 Pin	J8 Pin	Other 1/
						R		Ground
						T		Ground
						U		Ground
						V		Ground
						W		Ground
						X		Ground
						Y		Ground
							C	Ground
							F	Ground
							R	Ground
							T	Ground
							U	Ground
							V	Ground
							W	Ground
							X	Ground
							Y	Ground

1/ All points marked ground in the other column are interconnected.
2/ All points in the same horizontal line are interconnected.

TABLE X. HNF-3 connectors.

Quantity	Description	Suggested source and part number
1	J1	ITT Cannon Electric 1/ KPT2E20-41PDV, or equal Burndy Corporation 2/ BTO2E2041PH145, or equal
1	J1	ITT Cannon Electric KPT2E20-41PWDV, or equal Burndy Corporation BTO2E2041PWH145, or equal
1	J3	ITT Cannon Electric KPT2E12-10PWDV, or equal
1	J4	ITT Cannon Electric KPT2E14-5PDV, or equal
2	J5, J6	ITT Cannon Electric KPT2E18-32PWDV, or equal
2	J7, J8	ITT Cannon Electric KPT2E18-32PXDV, or equal

1/ FSCM 71468
2/ FSCM 09922

TABLE XI. HNF-3 mating connectors and backshells.

Quantity	Description	Suggested source and part number
1	Backshell	Glenair, Incorporated 1/ 390DS006NF1210A6, or equal
1	Backshell	Glenair, Incorporated 390DS006NF1410A6, or equal
2	Backshell	Glenair, Incorporated 390DS006NF2012A6, or equal

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TABLE XI. HNF-3 mating connectors and backshells. (Continued)

Quantity	Description	Suggested source and part number
1	Mating connector	MS3126E-20-41SW
1	Mating connector	MS3126E-12-10SW
1	Mating connector	MS3126E-14-5S
1	Mating connector	MS3126E-20-41S

1/ FSCM 06324

TABLE XII. MK-1955/USC interconnections, cable type IA.

P1	P2	P3 1/
	e	P
	S	U
	Z	J
	G	M
	J	E
	c	F
	K	B
	a	S
B	E	
E	A	
D	B	
H	T	
A	D	
C	C	

1/ All points in the same horizontal line are interconnected.

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TABLE XIII. MK-1955/USC interconnections, cable type IB.

P1	P2
A	S 1/
e	L
S	N
Y	B
G	C
C	F
N	H
P	e
L	W
M	J
D	X

1/ All points on the same horizontal line are interconnected.

TABLE XIV. MK-1955/USC connectors.

Quantity	Description	Part number
	Cable assembly type IA connectors	
1	P1	MS3126E12-10S
1	P2	MS3126E18-32SX
1	P3	MS3126E14-19S
	Cable assembly type IB connectors	
1	P1	MS3126E18-32SW
1	P2	MS3126E16-26S

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TABLE XV. First article inspection.

Examination or test	Requirement paragraph	Test paragraph
Dimensions and weight	3.2.4	4.6.5
Parts, materials, and processes	3.3	4.6.5
Finish	3.2.1.2	4.6.5
Identification and marking	3.4	4.5
Safety	3.2.3	4.7.1
General workmanship	3.5.1	4.7.2
Workmanship screen	3.5.2	4.6.2
TEMPEST	3.2.2.1.1.3	4.6.3.1
Temperature	3.2.2.1.2.1	4.6.3.2
Vibration	3.2.2.1.2.2	4.6.3.3
Shock	3.2.2.1.2.3	4.6.3.4
Humidity	3.2.2.1.2.4	4.6.3.5
Salt fog	3.2.2.1.2.5	4.6.3.6
Fungus	3.2.2.1.2.6	4.6.4
Enclosure	3.2.2.2	
Interchangeability	3.1, 3.1.2, 3.2.1, 3.2.1.1, 3.2.1.2.1, 3.2.1.2.2, 3.2.1.2.3, 3.2.1.3, 3.2.1.4, 3.2.1.5, 3.2.1.6, 3.2.1.7, 3.2.1.8, 3.2.1.9, 3.2.1.10, 3.2.1.11, 3.2.1.12.1, 3.2.1.12.2, 3.2.1.12.3, 3.2.1.12.4	
Power tests	3.2.2.1.1.1	4.6.1.1 4.6.6

TABLE XVI. Production inspection (Group A).

Examination or test	Requirement paragraph	Test paragraph	HNF-1 SSA	HNF-2A SSA	HNF-3-1 SSA	KG-84 SSA
General workmanship	3.5.1	4.7.1	X	X	X	X
Connectivity test	3.1	4.6.1.2	X	X	X	
Workmanship screen	3.5.2	4.7.2	X	X	X	X
Parts, materials, and processes	3.3	4.6.5	X	X	X	X
Finish	3.2.1.2	4.6.5	X	X	X	X
Identification and marking	3.4	4.6.5	X	X	X	X
Safety	3.2.3	4.5	X	X	X	X
Dimensions and weight	3.2.4	4.6.5	X	X	X	X

TABLE XVII. Sample composition.

Sample size	Sample composition 1/
2	1 HNF-1 SSA 1 HNF-2A SSA
2	1 HNF-3-1 SSA 1 KG-84 SSA
3	1 HNF-1 SSA 1 HNF-2A SSA 1 HNF-3-1 SSA

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TABLE XVII. Sample composition. (Continued)

Sample size	Sample composition 1/
3	1 KG-84 SSA 1 HNF-3-1 SSA 1 HNF-2A SSA
5	1 KG-84 SSA 1 HNF-3-1 SSA 1 HNF-2A SSA 1 HNF-1 SSA 1 KG-84 SSA
8	2 KG-84 SSA 2 HNF-3-1 SSA 2 HNF-2A SSA 2 HNF-1 SSA

1/ Sample compositions shall be chosen for a given sample size so that each SSA is tested at least once during every two month period. This is the reason there are multiple sample compositions for sample sizes two and three. If it is impossible to provide a sample composition shown, sample composition shall be decided by the Government QAR.

TABLE XVIII. Environmental inspection (Group C).

Examination or test	Requirement paragraph	Test paragraph
TEMPEST	3.2.2.1.1.3	4.6.2
Interchangeability	3.1, 3.1.2, 3.2.1, 3.2.1.1, 3.2.1.2.1, 3.2.1.2.2, 3.2.1.2.3, 3.2.1.3, 3.2.1.4, 3.2.1.5, 3.2.1.6, 3.2.1.7, 3.2.1.8, 3.2.1.9, 3.2.1.10, 3.2.1.11, 3.2.1.12.1, 3.2.1.12.2, 3.2.1.12.3, 3.2.1.12.4	4.6.1.1

TABLE XIX. Interchangeability tests.

Test to be performed	Production SSA under test.			
	KG-84 SSA	HNF-1 SSA	HNF-2A SSA	HNF-3-1 SSA
1.	X	X	X	X
2.	X	X	X	X
3.	X	X	X	X
4.		X	X	
5..	X	X	X	X
6.				X
7.		X		
8.			X	

1. Perform the connectivity test specified in 4.6.1.2 on the production SSA. Also verify that production interconnection box or back panel assembly can be exchanged with the interconnection box or back panel assembly on the Government provided SSA.

2. Verify that the production SSA can be mounted in a cabinet built according to MIL-C-28794.

3. Demonstrate that populating equipments slide into the mounting positions in the production SSA in the same manner that the populating equipments slide into the mounting position in the sample SSA provided by the Government

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4. Demonstrate that connectors on equipment populating the production SSA under test mate in the same manner as populating equipment connectors and connectors on the sample SSA provided by the Government.

5. Demonstrate that ATR retainers on the production SSA are capable of securing the populating equipments into their mounting positions in the same manner as ATR retainers on the sample SSA provided by the Government. Also verify that production ATR retainers can be exchanged with ATR retainers on the sample SSA provided by the Government.

6. Demonstrate that the Z-AKR/TSEC RCU can be mounted to the production HNF-3-1 SSA, in either position, in the same manner as it can be mounted to the sample HNF-3-1 SSA provided by the Government. This demonstration shall also include connecting the mounted RCU to the mounted TSEC/KY-75 using the cable provided by the Government. The cable shall be connected between J3 of the TSEC/KY-75 and J2 of the Z-AKR/TSEC. Verify that the production RCU bracket assembly can be exchanged with the RCU bracket assembly on the SSA provided by the Government.

7. Show that pin designations for connectors J7, J8, J10 and J11 on the production HNF-1 SSA are the same as pin designations for the same named connector on the sample HNF-1 SSA provided by the Government.

8. Shown that pin designations for connectors J7, J8, and J9 on the production HNF-2A SSA are the same as pin designations for connectors J7, J8, and J9, on the sample HNF-2A SSA provided by the Government.

TABLE XX. Connector interchangeability 1.

Mating connector 1/, 2/	SSA connector 1/, 3/
MS3126E-18-32S	J1 of the HNF-1 SSA
MS3126E-20-41SY	J2 of the HNF-1 SSA
MS3126E-14-5S	J4 of the HNF-1 SSA
MS3126E-22-55S	J1 of the HNF-2A SSA
MS3126E-24-61S	J2 of the HNF-2A SSA
MS3126E-14-19SX	J3 of the HNF-2A SSA
MS3126E-14-5S	J4 of the HNF-2A SSA
MS3126E-20-41S	J1 of the HNF-3-1 SSA
MS3126E-20-41SW	J2 of the HNF-3-1 SSA
MS3126E-12-10SW	J3 of the HNF-3-1 SSA
MS3126E-14-5S	J4 of the HNF-3-1 SSA

1/ The SSA connector and the corresponding mating connector are on the same horizontal line.

2/ Attach the mating connector to the corresponding SSA connector for each case. In each case the mating connector and the corresponding SSA connector shall fit together smoothly, and after the two are placed together a firm mechanical bond shall exist between the two connectors.

3/ For each case specified, ensure that the SSA connector is keyed the same as the mating connector. Pin designations shall be such that when the mating connector and SSA connector are attached, the pin designation on the SSA connector shall match the pin designations on the mating connector.

TABLE XXI. Connector interchangeability 2.

MK-1955/USC connector 1/, 2/	SSA connector 1/, 3/
P1 of cable assembly type IB	J5 of the HNF-3-1 SSA
P2 of cable assembly type IA	J7 of the HNF-3-1 SSA
P1 of cable assembly type IB	J6 of the HNF-3-1 SSA
P2 of cable assembly type IA	J8 of the HNF-3-1 SSA

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1/ The production SSA connector and the corresponding MK-1955/USC mating connector are on the same horizontal line.

2/ Attach the MK-1955/USC connector to the corresponding production SSA connector, for each case. In each case the MK-1955/USC connector and the corresponding production SSA connector shall fit together smoothly and after the two are placed together, a firm mechanical bond shall exist between the two connectors.

3/ For each case, ensure that the production SSA connector is keyed the same as the corresponding MK-1955/USC connector. Pin designations shall be such that when the production SSA connector and the corresponding MK-1955/USC connector are attached, the pin designations on the production SSA connector shall match those on the corresponding MK-1955/USC connector.

TABLE XXII. Connector interchangeability 3.

MK-1955/USC connector 1/, 2/	TSEC/KY-75 connector 1/, 3/
P2 of cable assembly type IB P3 of cable assembly type IA P1 of cable assembly type IA	J1 J2 J4

1/ The MK-1955/USC connector and corresponding TSEC/KY-75 connector are on the same horizontal line.

2/ Attach the production MK-1955/USC connector to the corresponding TSEC/KY-75 connector for each case. In each case the production MK-1955/USC connector and the corresponding TSEC/KY-75 connector shall fit together smoothly, and after the two are placed together, a firm mechanical bond shall exist between the two connectors.

3/ Ensure that the connectors on the production MK-1955/USC and the connectors on the sample MK-1955/USC are keyed in the same way.

TABLE XXIII. Connector interchangeability 4.

MK-1955/USC connector 1/, 2/	SSA connector 1/, 2/
P1 of cable assembly type IB P2 of cable assembly type IA P1 of cable assembly type IB P2 of cable assembly type IA	J5 of the HNF-3-1 SSA J7 of the HNF-3-1 SSA J6 of the HNF-3-1 SSA J8 of the HNF-3-1 SSA

1/ The HNF-3-1 SSA connector and the corresponding production MK-1955/USC mating connector are on the same horizontal line.

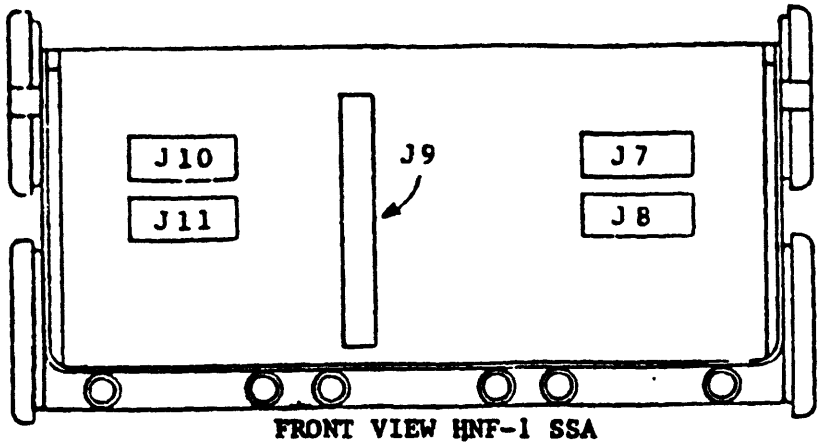
2/ Attach the production MK-1955/USC connector to the corresponding HNF-3-1 SSA connector for each case. In each case the production MK-1955/USC connector and the corresponding HNF-3-1 SSA shall fit together smoothly, and after the two are placed together a firm mechanical bond shall exist between the two connectors.

3/ For each case ensure that the production MK-1955/USC connector is keyed the same as the corresponding HNF-3-1 SSA connector. Pin designations shall be such that when the production MK-1955/USC connector and the corresponding HNF-3-1 SSA connector are attached, the pin designations on the production MK-1955/USC shall match pin designations on the corresponding HNF-3-1 SSA connector.

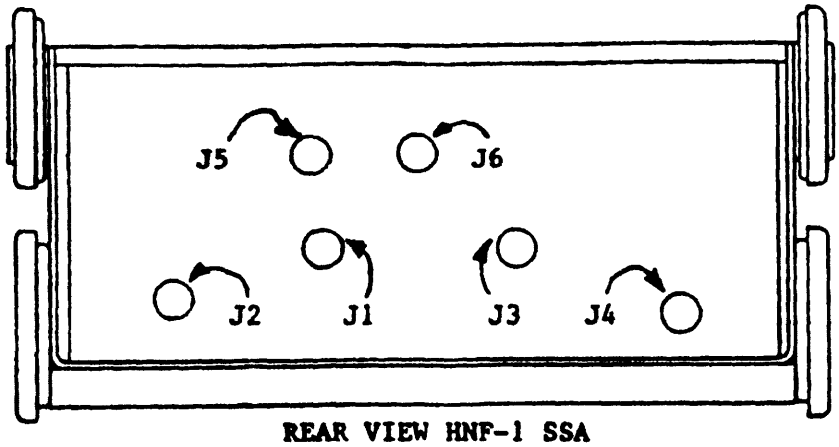
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E1	E2
E3	E4
E5	E6

FIGURE 1. Arrangement of E1, E2, E3, E4, E5, and E6.



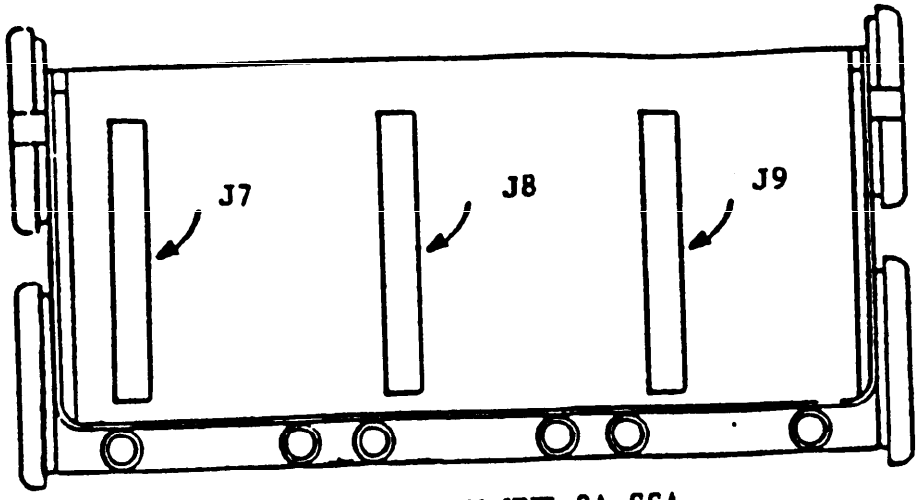
FRONT VIEW HNF-1 SSA



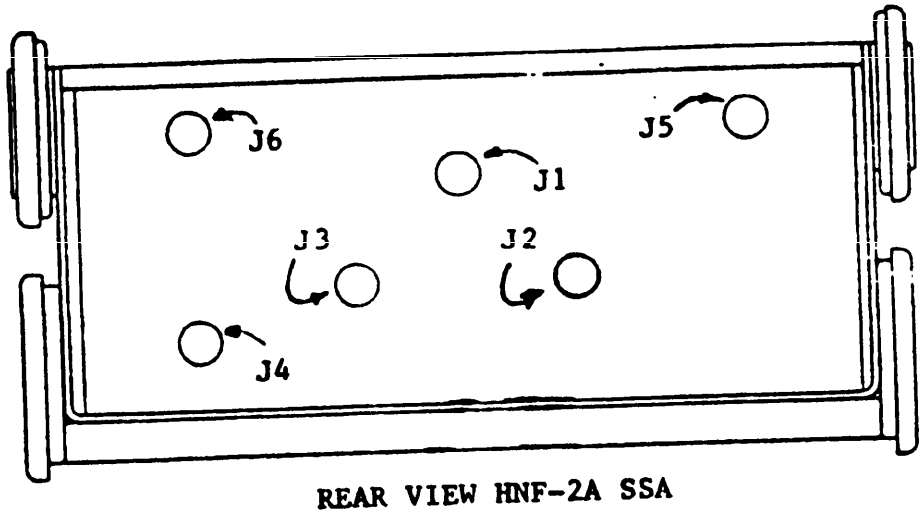
REAR VIEW HNF-1 SSA

FIGURE 2. HNF-1 connector locations.

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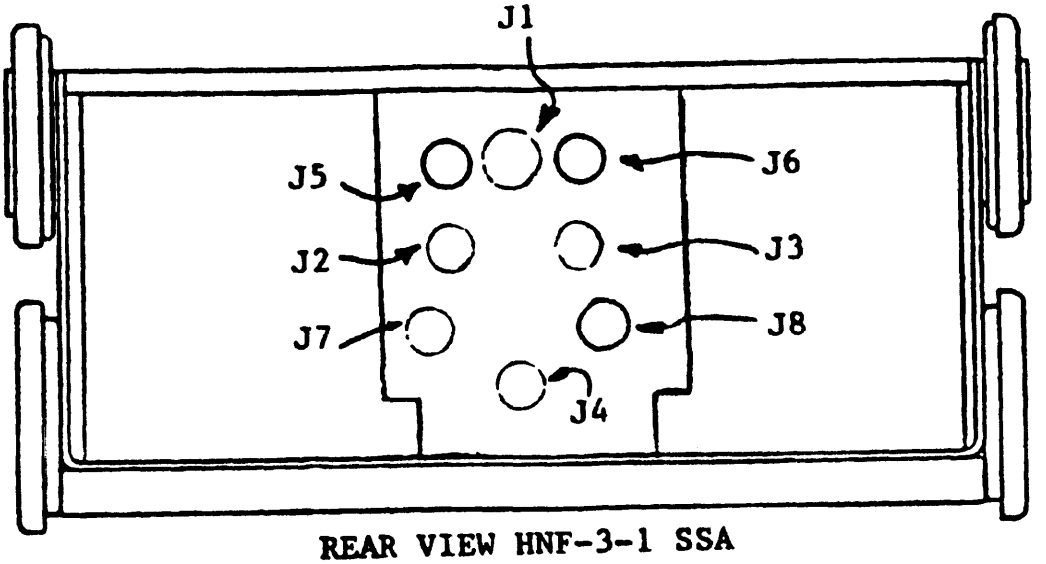
FRONT VIEW HNF-2A SSA



REAR VIEW HNF-2A SSA

FIGURE 3. HNF-2A connector locations.

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REAR VIEW HNF-3-1 SSA

FRONT VIEW OMITTED INTENTIONALLY
(THE FRONT HAS NO CONNECTORS)

FIGURE 4. HNF-3 connector locations.

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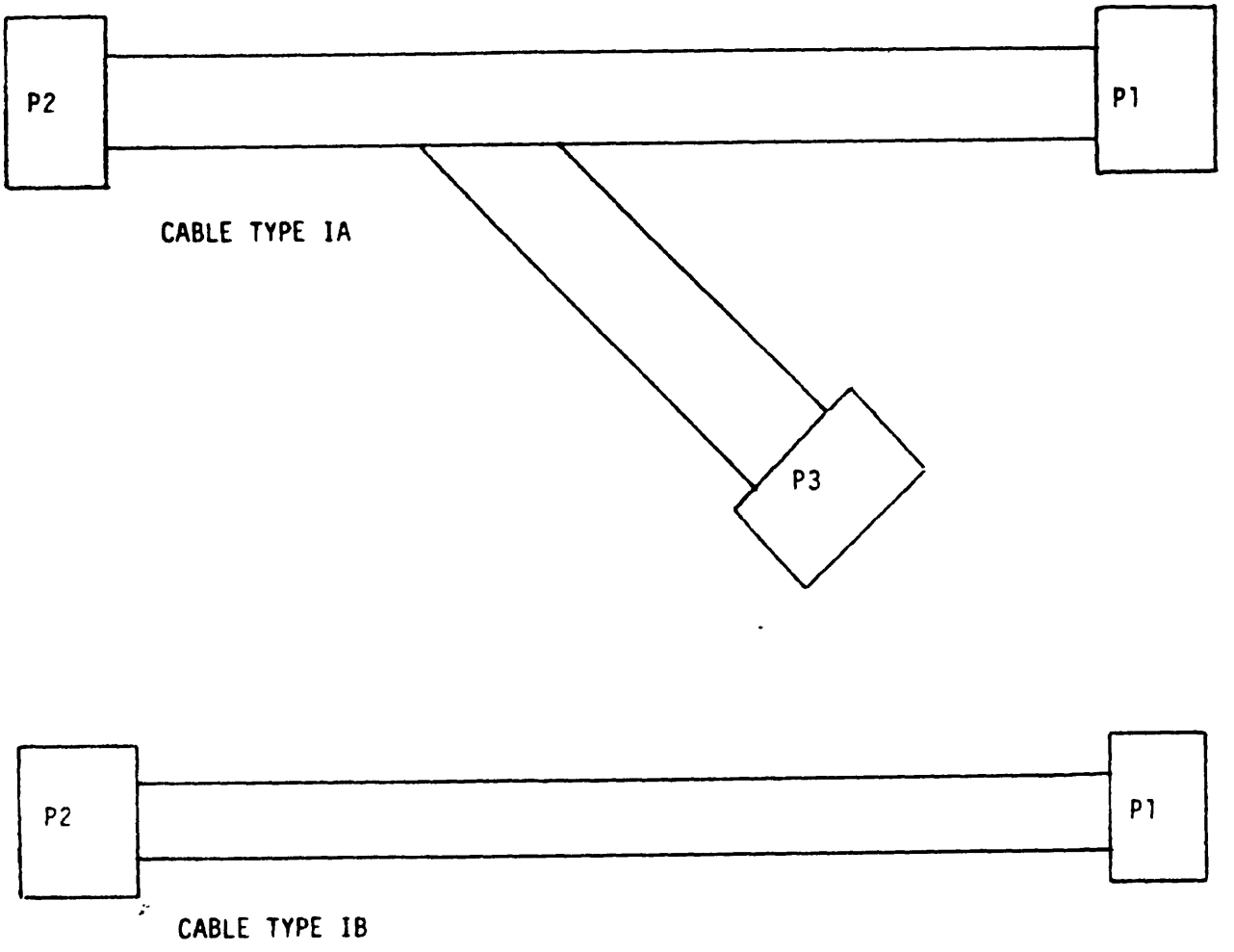
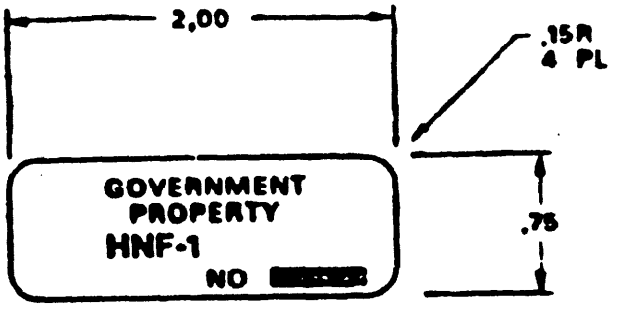


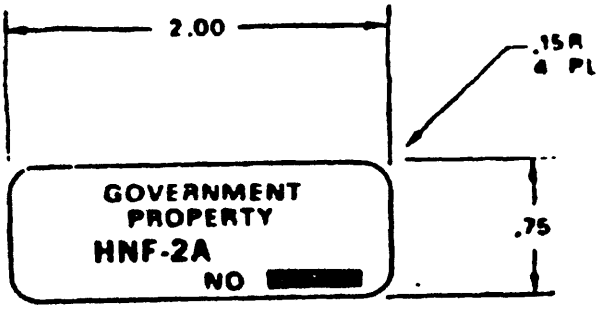
FIGURE 5. MK-1955/USC connector locations.

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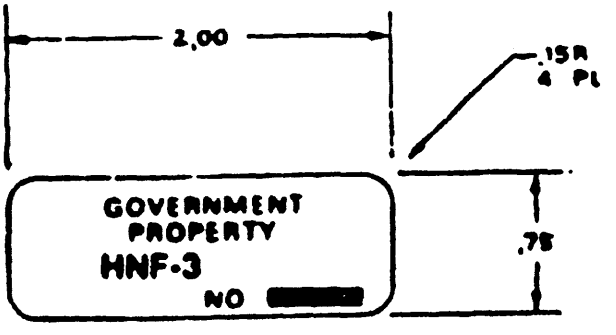
DIMENSIONS SHOWN ARE IN INCHES. ALL TOLERANCES ARE ±0.02



HNF-1 Nameplate



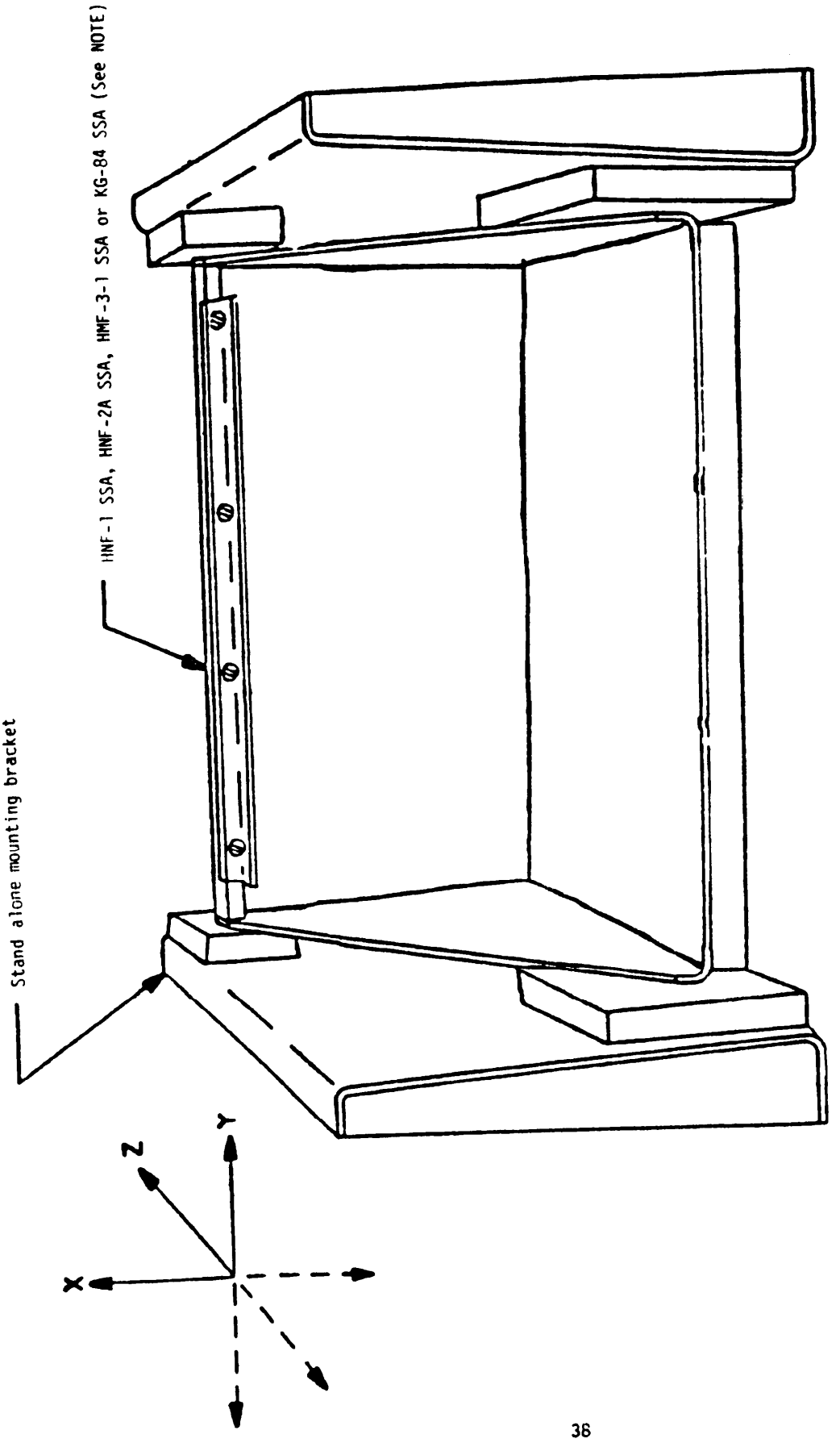
HNF-2A Nameplate



HNF-3 Nameplate

FIGURE 6. Nameplates.

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NOTE: SSA detail not shown.

FIGURE 7. Stand alone mount.

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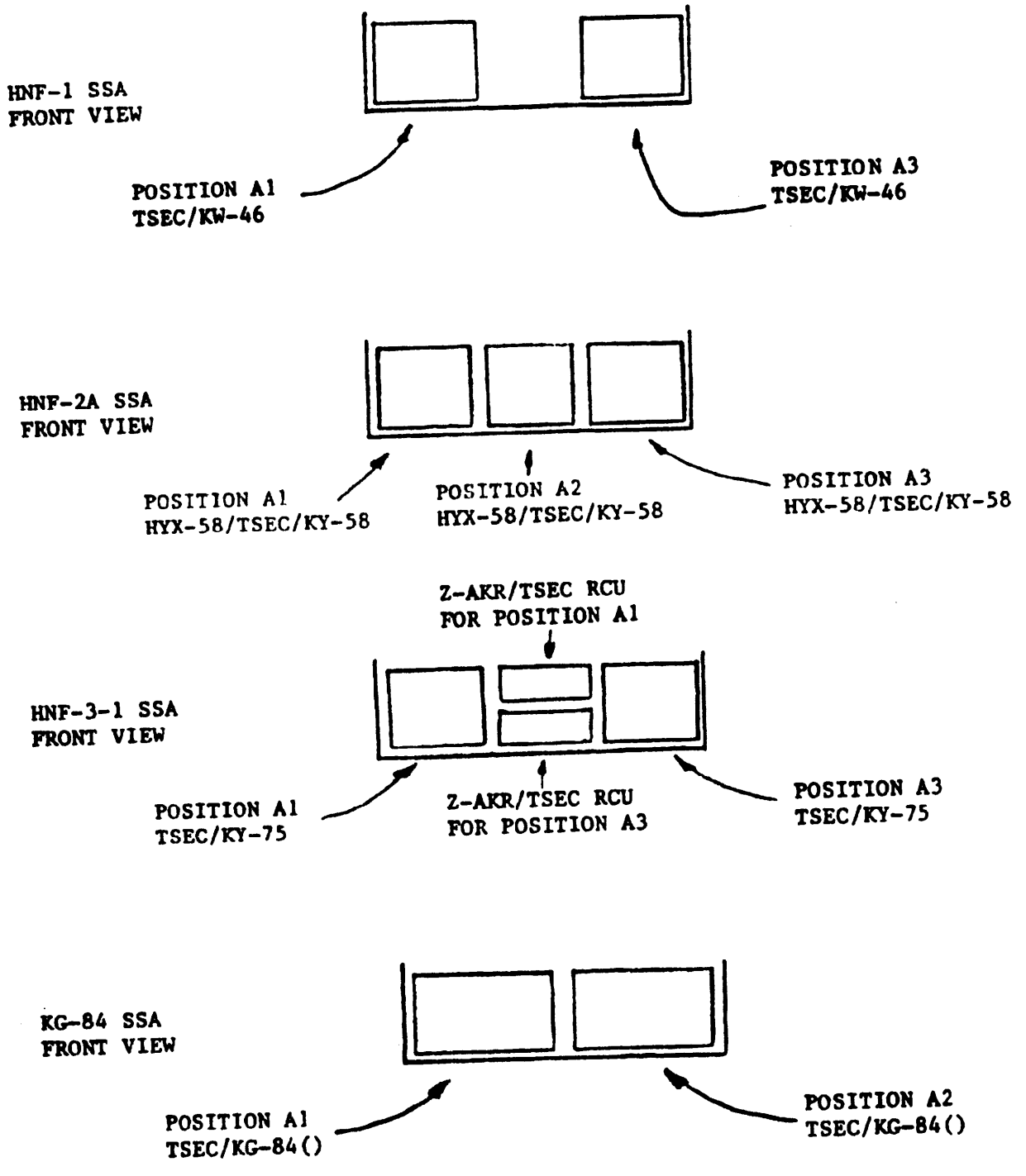


FIGURE 8. SSA populations.

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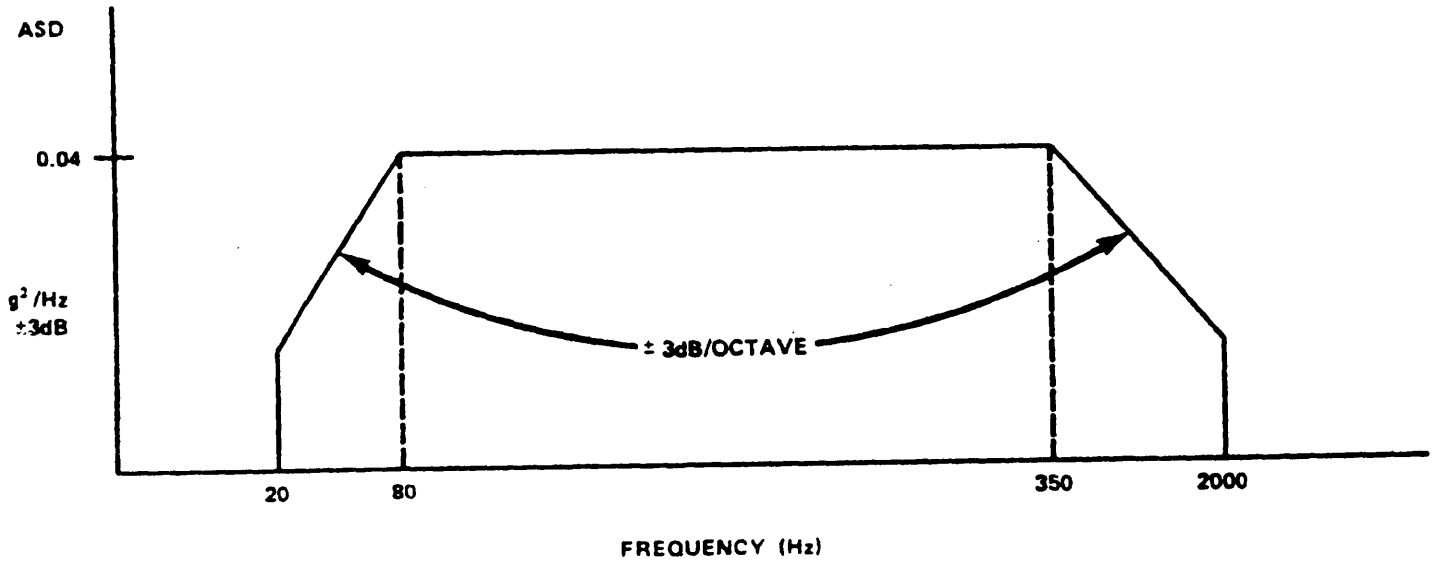


FIGURE 9. Random vibration curve.

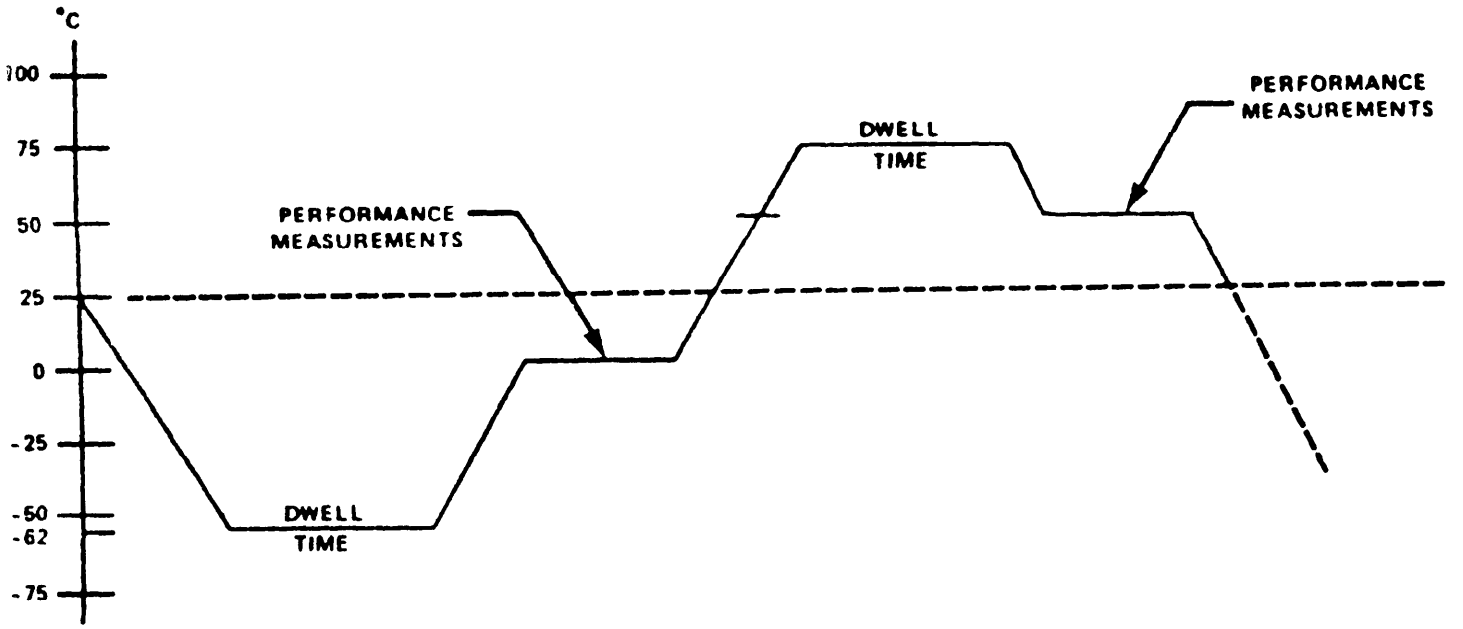


FIGURE 10. Temperature curve (one cycle).

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